

Nagaraj Mangilipalli

Data Engineer

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Professional Summary

Data Engineer with 5+ years of hands-on experience in designing, developing, and optimizing large-scale ETL/ELT pipelines, data lakes, and cloud-native data warehouse solutions across financial, healthcare, and enterprise domains. Experienced in building robust data ecosystems using Apache Spark, PySpark, Databricks, Snowflake, DBT, Airflow, and Kafka, ensuring reliable data ingestion, transformation, and orchestration for analytics and reporting. Proficient in leveraging AWS, Azure, and GCP platforms to architect scalable, high-performance, and cost-efficient data infrastructures. Demonstrated success in data modeling, performance tuning, and real-time streaming that enhance business decision-making and reduce operational inefficiencies. Adept at working with cross-functional teams including data scientists, analysts, and stakeholders to deliver secure, compliant, and insight-driven data solutions adhering to HIPAA, GDPR, and SOX regulatory frameworks.

Professional Experience

Data Engineer | Humana | USA

Jan 2025 – Current

- Designed and deployed end-to-end ETL pipelines using Apache Spark, Airflow, and DBT, automating ingestion from ERP, CRM, and IoT systems into Snowflake; this reduced refresh cycles by 40% and provided faster access to business-critical reports.
- Built and scaled real-time data streaming frameworks with Apache Kafka and AWS Kinesis, enabling ingestion of 200K+ messages per second with low latency, which supported fraud detection, operational dashboards, and customer engagement analytics.
- Led the migration of on-premises Teradata and Oracle data warehouses to Azure Synapse and Snowflake, applying schema redesign, clustering, and partitioning strategies that reduced query latency by 35% and cut storage costs.
- Partnered with data scientists to create feature engineering workflows in Databricks, integrating ML outputs back into curated Snowflake tables, shortening ML retraining cycles by 20% and boosting predictive accuracy.
- Established data quality and governance frameworks using SQL checks, Great Expectations, and lineage tracking, detecting schema drift, nulls, and duplicates, which improved trust and accuracy in downstream BI dashboards by 30%.
- Conducted performance tuning on PySpark jobs, SQL queries, and partitioning strategies, which reduced compute costs across cloud environments and improved runtime efficiency by 25%.
- Designed business-ready data marts and semantic layers in Snowflake aligned with financial and operational KPIs, enabling 1,000+ stakeholders to access standardized reports and reducing redundant reporting efforts by 40%.
- Automated deployment pipelines for DBT models and Spark jobs via Jenkins, Docker, and GitHub Actions, improving reliability, cutting release effort by 50%, and ensuring continuous delivery across environments.
- Collaborated in Agile sprints with product owners, testers, and architects, translating business requirements into technical deliverables, ensuring backlog grooming, and consistently achieving sprint goals.
- Applied GDPR, HIPAA, and SOX compliance measures including encryption, masking, and RBAC in cloud platforms, ensuring secure handling of customer and healthcare datasets and passing multiple audits without findings.

Data Engineer | Mphasis | India

Aug 2018 – July 2023

- Developed and optimized large-scale ETL pipelines using Apache Spark, Airflow, and Azure Data Factory to ingest, transform, and load multi-terabyte datasets into Snowflake and Delta Lake, reducing latency and improving data freshness by 40%.
- Designed and implemented data models and schemas in Snowflake, Redshift, and PostgreSQL, ensuring normalization, scalability, and faster query performance for analytics and reporting teams.
- Automated data workflows with Airflow, DBT, and Azure Data Factory, streamlining ingestion into Snowflake and accelerating reporting cycles, reducing discrepancies by 35% across global data warehouses.
- Implemented test automation frameworks with JUnit, Postman, and Selenium, achieving 90% coverage, reducing QA cycles by 35%, and significantly cutting defect leakage into production.
- Configured DevOps pipelines with Jenkins, Docker, and Kubernetes, automating containerized deployments, reducing manual deployment effort by 45%, and increasing release velocity.
- Designed dynamic and reusable front-end components with React and Angular, collaborating with UX teams to deliver consistent, accessible, and responsive user interfaces for global clients.
- Applied secure authentication methods (OAuth2, JWT, TLS) and encryption standards to APIs, ensuring compliance with GDPR, SOX, and industry-specific financial regulations.
- Built Tableau and Power BI dashboards to track real-time KPIs, system performance, and customer engagement, providing actionable insights to product managers and senior leadership.
- Partnered with BAs, QA engineers, and product managers in Agile ceremonies, ensuring alignment between business requirements and technical deliverables, increasing sprint velocity and reducing backlog issues.

Technical Skills

- Programming & Scripting:** Python (Pandas, PySpark, NumPy), SQL, Scala, R, Shell scripting
- Big Data & Processing:** Apache Spark, PySpark, Databricks, Hadoop, Flink, Kafka, Kinesis
- Data Warehousing:** Snowflake, Redshift, BigQuery, Azure Synapse, Teradata, Oracle, SQL Server
- ETL/Orchestration:** Apache Airflow, DBT, Informatica, Talend, Azure Data Factory, AWS Glue
- Cloud Platforms:** AWS (S3, Redshift, Lambda, SageMaker), Azure (Data Lake, Synapse, Databricks), GCP (BigQuery, Dataflow, Vertex AI)
- Databases:** PostgreSQL, MySQL, MongoDB, Cassandra, DynamoDB
- Data Modeling:** Star/Snowflake schema, Data Vault, Dimensional modeling, ER modeling
- DevOps & CI/CD:** Jenkins, GitHub Actions, Docker, Kubernetes, Terraform, MLflow
- Visualization & BI:** Tableau, Power BI, Looker, Grafana, Superset
- Security & Governance:** IAM, RBAC, PII/PHI masking, GDPR, HIPAA, SOX compliance

Education

- Masters** | University of Hartford, Hartford, CT, USA
- Bachelors** | Vardhaman College of Engineering, Hyderabad, Telangana, India.
- Aug 2023 – May 2025**
- Aug 2015 - May 2019**